My Solutions to Problem Sheet 3 QFT by David Tong

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1.

We need to show that $\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbf{I}$.

First, let $\mu = 0$ and $\nu = i$, then:

$$\{\gamma^0, \gamma^i\} = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \begin{bmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{bmatrix} + \begin{bmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{bmatrix} \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} = \begin{bmatrix} -\sigma^i & 0 \\ 0 & \sigma^i \end{bmatrix} + \begin{bmatrix} \sigma^i & 0 \\ 0 & -\sigma^i \end{bmatrix} = 0 \tag{1}$$

Let $\sigma^0 \equiv \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} = 1$, then $(\sigma^\mu)^2 = 1$, $\forall \mu = 0, 1, 2, 3$:

$$\begin{aligned} \{\gamma^0, \gamma^0\} &= 2(\gamma^0)^2 = 2 \begin{bmatrix} (\sigma^0)^2 & 0 \\ 0 & (\sigma^0)^2 \end{bmatrix} = 2\mathbf{I} \\ \{\gamma^i, \gamma^i\} &= 2(\gamma^i)^2 = 2 \begin{bmatrix} -(\sigma^i)^2 & 0 \\ 0 & -(\sigma^i)^2 \end{bmatrix} = -2\mathbf{I} \end{aligned} \tag{2}$$

Now, if $i \neq j$:

$$\begin{aligned} \{\gamma^i, \gamma^j\} = \begin{bmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{bmatrix} \begin{bmatrix} 0 & \sigma^j \\ -\sigma^j & 0 \end{bmatrix} + \begin{bmatrix} 0 & \sigma^j \\ -\sigma^j & 0 \end{bmatrix} \begin{bmatrix} 0 & \sigma^i \\ -\sigma^i & 0 \end{bmatrix} = \begin{bmatrix} -i\epsilon_k^{ij}\sigma^k & 0 \\ 0 & -i\epsilon_k^{ij}\sigma^k \end{bmatrix} \\ + \begin{bmatrix} -i\epsilon_k^{ji}\sigma^k & 0 \\ 0 & -i\epsilon_k^{ji}\sigma^k \end{bmatrix} = 0 \end{aligned} \tag{3}$$

So basically:

$$\{\gamma^\mu, \gamma^\nu\} = 2\eta^{\mu\nu}\mathbf{I}$$

From the looks of it, the γ^0 has been diagonalized using the U. The γ^0 has the following four eigen vectors, two degenerate eigen values, ± 1:

$$u_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ 1 \\ 0 \end{bmatrix}, \; u_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \; u_3 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 0 \\ -1 \\ 0 \end{bmatrix}, \; u_4 = \frac{1}{\sqrt{2}} \begin{bmatrix} 0 \\ 1 \\ 0 \\ -1 \end{bmatrix} \tag{4}$$

Therefore the speculated unitary looks like:

$$U = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \tag{5}$$

To confirm:

$$(\gamma')^0 = U\gamma^0 U^\dagger = \begin{bmatrix} 1 & 0 \\ 0 & -1 \end{bmatrix} \tag{6}$$

With a simple matrix multiplication one can also see that:

$$(\gamma')^i = U\gamma^i U^\dagger$$

2.

Assume the Clifford algebra

$$\{\gamma^\mu, \gamma^\nu\} = \gamma^\mu\gamma^\nu + \gamma^\nu\gamma^\mu = 2\eta^{\mu\nu}\mathbf{1}. \tag{7}$$

We first prove

$$\left[\gamma^\kappa\gamma^\lambda, \gamma^\mu\gamma^\nu\right] = 2\eta^{\lambda\mu}\gamma^\kappa\gamma^\nu - 2\eta^{\kappa\mu}\gamma^\lambda\gamma^\nu + 2\eta^{\lambda\nu}\gamma^\mu\gamma^\kappa - 2\eta^{\kappa\nu}\gamma^\mu\gamma^\lambda. \tag{8}$$

We use the identity

$$[AB, CD] = \{B, C\}AD - \{A, C\}BD + CA\{B, D\} - C\{A, D\}B, \tag{9}$$

which follows by repeated addition and subtraction of intermediate terms. Taking

$$A = \gamma^\kappa, \qquad B = \gamma^\lambda, \qquad C = \gamma^\mu, \qquad D = \gamma^\nu, \tag{10}$$

we get

$$\begin{aligned}[\gamma^\kappa\gamma^\lambda, \gamma^\mu\gamma^\nu] = {} & \{\gamma^\lambda, \gamma^\mu\}\gamma^\kappa\gamma^\nu - \{\gamma^\kappa, \gamma^\mu\}\gamma^\lambda\gamma^\nu \\ & + \gamma^\mu\gamma^\kappa\{\gamma^\lambda, \gamma^\nu\} - \gamma^\mu\{\gamma^\kappa, \gamma^\nu\}\gamma^\lambda.\end{aligned} \tag{11}$$

Using

$$\{\gamma^\alpha, \gamma^\beta\} = 2\eta^{\alpha\beta}\mathbf{1}, \tag{12}$$

this becomes

$$\begin{aligned}[\gamma^\kappa\gamma^\lambda, \gamma^\mu\gamma^\nu] = {} & 2\eta^{\lambda\mu}\gamma^\kappa\gamma^\nu - 2\eta^{\kappa\mu}\gamma^\lambda\gamma^\nu \\ & + 2\eta^{\lambda\nu}\gamma^\mu\gamma^\kappa - 2\eta^{\kappa\nu}\gamma^\mu\gamma^\lambda.\end{aligned} \tag{13}$$

This proves the required identity.

Now define

$$S^{\mu\nu} \equiv \frac{1}{4}[\gamma^\mu, \gamma^\nu]. \tag{14}$$

Using the Clifford algebra,

$$\gamma^\nu\gamma^\mu = 2\eta^{\mu\nu} - \gamma^\mu\gamma^\nu, \tag{15}$$

we find

$$\begin{aligned} S^{\mu\nu} &= \frac{1}{4}\left(\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu\right) \\ &= \frac{1}{4}\left(\gamma^\mu\gamma^\nu - 2\eta^{\mu\nu} + \gamma^\mu\gamma^\nu\right) \\ &= \frac{1}{2}\left(\gamma^\mu\gamma^\nu - \eta^{\mu\nu}\right). \end{aligned} \tag{16}$$

Finally, we show that the matrices $S^{\mu\nu}$ form a representation of the Lorentz Lie algebra. Since the metric terms commute with everything,

$$[S^{\mu\nu}, S^{\rho\sigma}] = \frac{1}{4}[\gamma^\mu\gamma^\nu, \gamma^\rho\gamma^\sigma]. \tag{17}$$

Using the identity proved above, with

$$\kappa = \mu, \qquad \lambda = \nu, \tag{18}$$

we obtain

$$\begin{aligned} [S^{\mu\nu}, S^{\rho\sigma}] &= \frac{1}{4}\left(2\eta^{\nu\rho}\gamma^\mu\gamma^\sigma - 2\eta^{\mu\rho}\gamma^\nu\gamma^\sigma + 2\eta^{\nu\sigma}\gamma^\rho\gamma^\mu - 2\eta^{\mu\sigma}\gamma^\rho\gamma^\nu\right) \\ &= \frac{1}{2}\eta^{\nu\rho}\gamma^\mu\gamma^\sigma - \frac{1}{2}\eta^{\mu\rho}\gamma^\nu\gamma^\sigma + \frac{1}{2}\eta^{\nu\sigma}\gamma^\rho\gamma^\mu - \frac{1}{2}\eta^{\mu\sigma}\gamma^\rho\gamma^\nu. \end{aligned} \tag{19}$$

Using

$$\frac{1}{2}\gamma^\alpha\gamma^\beta = S^{\alpha\beta} + \frac{1}{2}\eta^{\alpha\beta}, \tag{20}$$

we get

$$[S^{\mu\nu}, S^{\rho\sigma}] = \eta^{\nu\rho}S^{\mu\sigma} - \eta^{\mu\rho}S^{\nu\sigma} + \eta^{\nu\sigma}S^{\rho\mu} - \eta^{\mu\sigma}S^{\rho\nu}. \tag{21}$$

Since

$$S^{\rho\mu} = -S^{\mu\rho}, \qquad S^{\rho\nu} = -S^{\nu\rho}, \tag{22}$$

this becomes

$$\boxed{[S^{\mu\nu}, S^{\rho\sigma}] = \eta^{\nu\rho}S^{\mu\sigma} - \eta^{\mu\rho}S^{\nu\sigma} + \eta^{\mu\sigma}S^{\nu\rho} - \eta^{\nu\sigma}S^{\mu\rho}}. \tag{23}$$

This is precisely the Lorentz Lie algebra in the spinor representation. Hence the matrices

$$S^{\mu\nu} = \frac{1}{4}[\gamma^\mu, \gamma^\nu] \tag{24}$$

form a representation of the Lie algebra of the Lorentz group.

3.

i.

In chiral representation, $\mathrm{Tr}\{\gamma^\mu\} = 0$, hence in any other, we use the cyclicity:

$$\mathrm{Tr}\{U\gamma^\mu U^\dagger\} = \mathrm{Tr}\{U^\dagger U\gamma^\mu\} = \mathrm{Tr}\{\gamma^\mu\} = 0 \tag{25}$$

ii.

$$\mathrm{Tr}\{\{\gamma^\mu,\gamma^\nu\}\} = 2\,\mathrm{Tr}\{\gamma^\mu\gamma^\nu\} = 8\eta^{\mu\nu} \implies \mathrm{Tr}\{\gamma^\mu\gamma^\nu\} = 4\eta^{\mu\nu} \tag{26}$$

iii.

preliminary iii.A:

$$\{\gamma^5,\gamma^\mu\} = 0$$

.

pf:

Let $\mu = 3$:

$$\{\gamma^5,\gamma^3\} = -i\{\gamma^0\gamma^1\gamma^2\gamma^3,\gamma^3\} = -i(\gamma^0\gamma^1\gamma^2\gamma^3\gamma^3 + \gamma^3\gamma^0\gamma^1\gamma^2\gamma^3)$$
$$= -i(\gamma^0\gamma^1\gamma^2\gamma^3\gamma^3 + (-1)^3\gamma^0\gamma^1\gamma^2\gamma^3\gamma^3) = 0$$

Let $\mu = 2$:

$$\{\gamma^5,\gamma^2\} = -i\{\gamma^0\gamma^1\gamma^2\gamma^3,\gamma^2\} = -i(\gamma^0\gamma^1\gamma^2\gamma^3\gamma^2 + \gamma^2\gamma^0\gamma^1\gamma^2\gamma^3)$$
$$= -i(-\gamma^0\gamma^1\gamma^2\gamma^2\gamma^3 + (-1)^2\gamma^0\gamma^1\gamma^2\gamma^2\gamma^3) = 0$$

$\mu = 1$ is similar, just out of sheer justification, for $\mu = 0$:

$$\{\gamma^5,\gamma^0\} = -i\{\gamma^0\gamma^1\gamma^2\gamma^3,\gamma^0\} = -i(\gamma^0\gamma^1\gamma^2\gamma^3\gamma^0 + \gamma^0\gamma^0\gamma^1\gamma^2\gamma^3)$$
$$= -i((-1)^3\gamma^0\gamma^0\gamma^1\gamma^2\gamma^3 + \gamma^0\gamma^0\gamma^1\gamma^2\gamma^3) = 0 \quad \blacksquare$$

With this in hand:

$$\mathrm{Tr}\{\gamma^\mu\gamma^\nu\gamma^\rho\} = \mathrm{Tr}\big\{(\gamma^5)^2\gamma^\mu\gamma^\nu\gamma^\rho\big\} = \mathrm{Tr}\big\{\gamma^5\gamma^\mu\gamma^\nu\gamma^\rho\gamma^5\big\} =$$

$$\mathrm{Tr}\big\{(-1)^3\gamma^\mu\gamma^\nu(\gamma^5)^2\gamma^\rho\big\} = -\,\mathrm{Tr}\{\gamma^\mu\gamma^\nu\gamma^\rho\}$$
$$\implies \mathrm{Tr}\{\gamma^\mu\gamma^\nu\gamma^\rho\} = 0. \quad \blacksquare$$

iv.

$$\begin{aligned}(\gamma^5)^2 &= -\gamma^0\gamma^1\gamma^2\gamma^3\gamma^0\gamma^1\gamma^2\gamma^3 = -(-1)^3(\gamma^0)^2\gamma^1\gamma^2\gamma^3\gamma^1\gamma^2\gamma^3 \\ &= \gamma^1\gamma^2\gamma^3\gamma^1\gamma^2\gamma^3 = (-1)^2(\gamma^1)^2\gamma^2\gamma^3\gamma^2\gamma^3 = -\gamma^2\gamma^3\gamma^2\gamma^3 \\ &= (\gamma^2)^2(\gamma^3)^2 = 1 \quad \blacksquare\end{aligned} \tag{27}$$

v.

$$\mathrm{Tr}\{\gamma^5\} = \mathrm{Tr}\{-i\gamma^0\gamma^1\gamma^2\gamma^3\} = i\,\mathrm{Tr}\{\gamma^1\gamma^0\gamma^2\gamma^3\} = i\,\mathrm{Tr}\{\gamma^1\gamma^2\gamma^3\gamma^0\} \tag{28}$$

But, also:

$$\mathrm{Tr}\{\gamma^5\} = \mathrm{Tr}\{-i\gamma^0\gamma^1\gamma^2\gamma^3\} = -i\,\mathrm{Tr}\{\gamma^1\gamma^2\gamma^3\gamma^0\} \tag{29}$$

Therefore:

$$\mathrm{Tr}\{\gamma^5\} = -\,\mathrm{Tr}\{\gamma^5\} \implies \mathrm{Tr}\{\gamma^5\} = 0. \quad \blacksquare \tag{30}$$

vi.

$$\not{p}\not{q} = p_\mu\gamma^\mu\, q_\nu\gamma_\nu = p_\mu q_\nu(2\eta^{\mu\nu} - \gamma^\nu\gamma^\mu) = 2\,p.q \;-\; \not{q}\not{p}. \quad \blacksquare \tag{31}$$

We know that:

$$4S^{\mu\nu} = \gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu \tag{32}$$

Hence:

$$\not{p}\not{q} = p_\mu q_\nu(\eta^{\mu\nu} + 2S^{\mu\nu}) = p\,.\,q \;+ 2S^{\mu\nu}p_\mu q_\nu. \quad \blacksquare \tag{33}$$

vii.

preliminary vii.A:

$$\mathrm{Tr}\{S^{\mu\nu}\} = 0$$

pf:

$$\mathrm{Tr}\{S^{\mu\nu}\} = \frac{1}{4}\,\mathrm{Tr}\{[\gamma^\mu,\gamma^\nu]\} = \frac{1}{4}\,\mathrm{Tr}\{\gamma^\mu\gamma^\nu - \gamma^\nu\gamma^\mu\} = \eta^{\mu\nu} - \eta^{\nu\mu} = 0. \quad \blacksquare \tag{34}$$

Knowing this and using (vi):

$$\mathrm{Tr}\{\not{p}\not{q}\} = \mathrm{Tr}\{p\,.\,q \;+ 2S^{\mu\nu}p_\mu q_\nu.\} = 4p.q\,. \quad \blacksquare \tag{35}$$

viii.

First, let's explore this for simpler cases of $n = 1$ and $n = 3$. Let $n = 1$:

$$\mathrm{Tr}\Big\{\not{p}_1\Big\} = \mathrm{Tr}\{\gamma^{\mu_1}p_{1\mu_1}\} = p_{1\mu_1}\,\mathrm{Tr}\{\gamma^{\mu_1}\} = 0. \quad \blacksquare \tag{36}$$

Let $n = 3$, by (iii):

$$\mathrm{Tr}\Big\{\not{p}_1\not{p}_2\not{p}_3\Big\} = p_{1\mu_1}p_{2\mu_2}p_{3\mu_3}\,\mathrm{Tr}\{\gamma^{\mu_1}\gamma^{\mu_2}\gamma^{\mu_3}\} = 0. \quad \blacksquare \tag{37}$$

Now let $n = 2l + 1$, for some $l \in \mathbb{N}$ and we use (iii.A) repeatedly:

$$\begin{aligned}\operatorname{Tr}\Big\{\not{p}_1 \dots \not{p}_n\Big\} &= p_{1\mu_1} \dots p_{n\mu_n} \operatorname{Tr}\{\gamma^5 \gamma^{\mu_1} \dots \gamma^{\mu_l} \gamma^{\mu_{l+1}} \dots \gamma^{\mu_{2l+1}} \gamma^5\} \\ &= p_{1\mu_1} \dots p_{n\mu_n} \operatorname{Tr}\Big\{\gamma^{\mu_1} \dots \gamma^{\mu_l} (-1)^l \gamma^5 \gamma^5 (-1)^{(l+1)} \gamma^{\mu_{l+1}} \\ &\dots \gamma^{\mu_{2l+1}}\} = -\operatorname{Tr}\Big\{\not{p}_1 \dots \not{p}_n\Big\} \\ &\implies \operatorname{Tr}\Big\{\not{p}_1 \dots \not{p}_n\Big\} = 0. \quad \blacksquare\end{aligned} \tag{38}$$

ix.

$$\begin{aligned}\operatorname{Tr}\Big\{\not{p}_1 \not{p}_2 \not{p}_3 \not{p}_4\Big\} &= \overbrace{p_{1\mu} p_{2\nu} p_{3\rho} p_{4\sigma}}^{A_{\mu\nu\rho\sigma}} \operatorname{Tr}\{\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma\} \\ &= A_{\mu\nu\rho\sigma} \operatorname{Tr}\{(2\eta^{\mu\nu} - \gamma^\nu \gamma^\mu)\gamma^\rho \gamma^\sigma\} \\ &= 2A_{\mu\nu\rho\sigma}\eta^{\mu\nu} \operatorname{Tr}\{\gamma^\rho \gamma^\sigma\} - A_{\mu\nu\rho\sigma} \operatorname{Tr}\{\gamma^\nu \gamma^\mu \gamma^\rho \gamma^\sigma\} \\ &= 8A_{\mu\nu\rho\sigma}\eta^{\mu\nu}\eta^{\rho\sigma} - A_{\mu\nu\rho\sigma} \operatorname{Tr}\{\gamma^\nu (2\eta^{\mu\rho} - \gamma^\rho \gamma^\mu)\gamma^\sigma\} \\ &= 8(p_1.p_2)(p_3.p_4) - 8(p_1.p_3)(p_2.p_4) \\ &+ 8(p_1.p_4)(p_2.p_3) - \operatorname{Tr}\Big\{\not{p}_1 \not{p}_2 \not{p}_3 \not{p}_4\Big\}\end{aligned} \tag{39}$$

Hence:

$$\begin{aligned}\operatorname{Tr}\Big\{\not{p}_1 \not{p}_2 \not{p}_3 \not{p}_4\Big\} &= 4(p_1.p_2)(p_3.p_4) - 4(p_1.p_3)(p_2.p_4) \\ &+ 4(p_1.p_4)(p_2.p_3)\,. \quad \blacksquare\end{aligned} \tag{40}$$

x.

$$\operatorname{Tr}\Big\{\gamma^5 \not{p}_1 \not{p}_2\Big\} = p_{1\mu} p_{2\nu} \operatorname{Tr}\{\gamma^5 \gamma^\mu \gamma^\nu\} \tag{41}$$

Let $A = \gamma^5 \gamma^\mu \gamma^\nu$, in 4 dimensional spacetime, there always exists index ρ different from chosen μ and ν, and hence:

$$\{\gamma^\rho, \gamma^5\} = \{\gamma^\rho, \gamma^\nu\} = \{\gamma^\rho, \gamma^\mu\} = 0 \tag{42}$$

Which means that:

$$\gamma^\rho A = -A\gamma^\rho \tag{43}$$

Since $(\gamma^\rho)^2 = \eta^{\rho\rho} I$, therefore γ^ρ has an inverse, and hence:

$$\gamma^\rho A(\gamma^\rho)^{-1} = -A \implies \operatorname{Tr}\{A\} = -\operatorname{Tr}\{A\} \implies \operatorname{Tr}\{A\} = 0. \tag{44}$$

Therefore:

$$\operatorname{Tr}\Big\{\gamma^5 \not{p}_1 \not{p}_2\Big\} = p_{1\mu} p_{2\nu} \operatorname{Tr}\{A\} = 0. \quad \blacksquare \tag{45}$$

xi.

$$\begin{aligned}\gamma_\mu \not{p} \gamma^\mu &= p_\nu \eta_{\sigma\mu} \gamma^\sigma \gamma^\nu \gamma^\mu = p_\nu \eta_{\sigma\mu}(2\eta^{\sigma\nu} - \gamma^\nu \gamma^\sigma)\gamma^\mu = 2\not{p} - \not{p}\gamma_\mu \gamma^\mu \\ &= -2\not{p}. \quad \blacksquare\end{aligned} \tag{46}$$

xii.

$$\begin{aligned}\gamma_\mu \not{p}_1 \not{p}_2 \gamma^\mu &= p_{2\nu} \gamma_\mu \not{p}_1 \gamma^\nu \gamma^\mu = p_{1\rho} p_{2\nu} \gamma_\mu \gamma^\rho (2\eta^{\mu\nu} - \gamma^\mu \gamma^\nu) \\ &= 2\not{p}_2 \not{p}_1 - \gamma_\mu \not{p}_1 \gamma^\mu \not{p}_2 = 2\{\not{p}_1, \not{p}_2\} = 4p_1.p_2 \quad \blacksquare\end{aligned} \tag{47}$$

xiii.

$$\begin{aligned}\gamma_\mu \not{p}_1 \not{p}_2 \not{p}_3 \gamma^\mu &= p_{3\nu} \gamma_\mu \not{p}_1 \not{p}_2 (2\eta^{\mu\nu} - \gamma^\mu \gamma^\nu) = 2\not{p}_3 \not{p}_1 \not{p}_2 \\ &- (\gamma_\mu \not{p}_1 \not{p}_2 \gamma^\mu)\not{p}_3 = 2\not{p}_3 \not{p}_1 \not{p}_2 - 2(2p_1.p_2)\not{p}_3 \\ &= 2\not{p}_3 \not{p}_1 \not{p}_2 - 4\not{p}_3 (p_1.p_2) = 2\not{p}_3 (\not{p}_1 \not{p}_2 - 2p_1.p_2) \\ &= -2\not{p}_3 \not{p}_2 \not{p}_1. \quad \blacksquare\end{aligned} \tag{48}$$

xiv.

$$\mathrm{Tr}\Big\{\gamma^5 \not{p}_1 \not{p}_2 \not{p}_3 \not{p}_4\Big\} = p_1^\mu p_2^\nu p_3^\rho p_4^\sigma \,\mathrm{Tr}\big\{\gamma^5 \gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma\big\} \tag{49}$$

If any pair of indices are equal we will get a term prop. to $\mathrm{Tr}\big\{\gamma^5 \gamma^\alpha \gamma^\beta\big\}$ which is 0 as was shown in (x). Therefore no two indices among $\{\mu, \nu, \rho, \sigma\}$ can be equal, and moreover, we are in 4-d spacetime, hence the above expression can be combinatorial of $\{0, 1, 2, 3\}$. Now let, for example, $\mu = 0, \nu = 1, \rho = 2, \sigma = 3$. Then we have that:

$$A^{\mu\nu\rho\sigma} \,\mathrm{Tr}\big\{-i\gamma^0 \gamma^1 \gamma^2 \gamma^3 \,\gamma^\mu \gamma^\nu \gamma^\rho \gamma^\sigma\big\} = A^{\mu\nu\rho\sigma} \,\mathrm{Tr}\{iI\} = 4iA^{\mu\nu\rho\sigma} \tag{50}$$

Any other combinatorial of this sequence is contributing to the original expression except for the fact that if two indices are exchanged, there will be a -1 sign as due to the fact that in our case $\{\gamma^\mu, \gamma^\nu\} = 0$. We can use a symbol that captures these sign changes, and that is the Levi-Civita symbol in 4-d. Hence:

$$\mathrm{Tr}\Big\{\gamma^5 \not{p}_1 \not{p}_2 \not{p}_3 \not{p}_4\Big\} = 4i\epsilon_{\mu\nu\rho\sigma} \,p_1^\mu p_2^\nu p_3^\rho p_4^\sigma. \quad \blacksquare \tag{51}$$

4.

We first prove the Eq.(5) set in the problems list. We have that:

$$\begin{aligned}u^r(\vec{p})^\dagger . u^s(\vec{p}) &= \begin{bmatrix}\sqrt{p.\sigma}(\xi^r)^\dagger & \sqrt{p.\bar{\sigma}}(\xi^r)^\dagger\end{bmatrix} \begin{bmatrix}\sqrt{p.\sigma}\xi^s \\ \sqrt{p.\bar{\sigma}}\xi^s\end{bmatrix} = p.\sigma \,\delta^{rs} + p.\bar{\sigma} \,\delta^{rs} \\ &= 2p_0 \delta^{rs}\end{aligned} \tag{52}$$

$$\begin{aligned}\bar{u}^r(\vec{p})\,.\,u^s(\vec{p}) &= \begin{bmatrix}\sqrt{p.\sigma}(\xi^r)^\dagger & \sqrt{p.\bar{\sigma}}(\xi^r)^\dagger\end{bmatrix}\gamma^0\begin{bmatrix}\sqrt{p.\sigma}\xi^s\\ \sqrt{p.\bar{\sigma}}\xi^s\end{bmatrix}\\ &= 2\sqrt{(p.\sigma)(p.\bar{\sigma})}\,\delta^{rs}\end{aligned} \tag{53}$$

We have that:

$$\begin{aligned}(p.\sigma)(p.\bar{\sigma}) &= (p_0+p_i\sigma^i)(p_0-p_i\sigma^i) = p_0^2 - p_ip_j\sigma^i\sigma^j\\ &= p_0^2 - p^ip^j(\delta_{ij}+i\epsilon_{ijk}\sigma^k) = m^2 - ip^ip^j\epsilon_{ijk}\sigma^k\\ &= m^2\end{aligned} \tag{54}$$

Hence:

$$\bar{u}^r(\vec{p})\,.\,u^s(\vec{p}) = 2m\delta^{rs}. \quad \blacksquare \tag{55}$$

Similarly for Eqs. (6):

$$\begin{aligned}v^r(\vec{p})^\dagger.v^s(\vec{p}) &= \begin{bmatrix}\sqrt{p.\sigma}(\xi^r)^\dagger & -\sqrt{p.\bar{\sigma}}(\xi^r)^\dagger\end{bmatrix}\begin{bmatrix}\sqrt{p.\sigma}\xi^s\\ -\sqrt{p.\bar{\sigma}}\xi^s\end{bmatrix} = p.\sigma\,\delta^{rs} + p.\bar{\sigma}\,\delta^{rs}\\ &= 2p_0\delta^{rs}\end{aligned} \tag{56}$$

$$\begin{aligned}\bar{v}^r(\vec{p})\,.\,v^s(\vec{p}) &= \begin{bmatrix}\sqrt{p.\sigma}(\xi^r)^\dagger & -\sqrt{p.\bar{\sigma}}(\xi^r)^\dagger\end{bmatrix}\gamma^0\begin{bmatrix}\sqrt{p.\sigma}\xi^s\\ -\sqrt{p.\bar{\sigma}}\xi^s\end{bmatrix}\\ &= -2\sqrt{(p.\sigma)(p.\bar{\sigma})}\,\delta^{rs} = -2m\delta^{rs}. \quad \blacksquare\end{aligned} \tag{57}$$

For Eq. (7):

$$\begin{aligned}\bar{u}^s(\vec{p})\,.\,v^r(\vec{p}) &= \begin{bmatrix}\sqrt{p.\sigma}(\xi^s)^\dagger & \sqrt{p.\bar{\sigma}}(\xi^s)^\dagger\end{bmatrix}\gamma^0\begin{bmatrix}\sqrt{p.\sigma}\xi^r\\ -\sqrt{p.\bar{\sigma}}\xi^r\end{bmatrix}\\ &= (\sqrt{p.\sigma}\sqrt{p.\bar{\sigma}} - \sqrt{p.\sigma}\sqrt{p.\bar{\sigma}})\delta^{sr} = 0. \quad \blacksquare\end{aligned} \tag{58}$$

And Finally Eq. (8):

$$\begin{aligned}(u^s(\vec{p}))^\dagger\,.\,v^r(-\vec{p}) &= \begin{bmatrix}\sqrt{p.\sigma}(\xi^s)^\dagger & \sqrt{p.\bar{\sigma}}(\xi^s)^\dagger\end{bmatrix}\begin{bmatrix}\sqrt{p.\bar{\sigma}}\xi^r\\ -\sqrt{p.\sigma}\xi^r\end{bmatrix}\\ &= (\sqrt{p.\sigma}\sqrt{p.\bar{\sigma}} - \sqrt{p.\sigma}\sqrt{p.\bar{\sigma}})\delta^{sr} = 0. \quad \blacksquare\end{aligned} \tag{59}$$

5.

For Eq. (9):

$$\begin{aligned}\sum_{s=1}^{2} u^s(\vec{p})\bar{u}^s(\vec{p}) &= \sum_{s=1}^{2}\begin{bmatrix}\sqrt{p.\sigma}\xi^s\\ \sqrt{p.\bar{\sigma}}\xi^s\end{bmatrix}\begin{bmatrix}\sqrt{p.\sigma}(\xi^s)^\dagger & \sqrt{p.\bar{\sigma}}(\xi^s)^\dagger\end{bmatrix}\gamma^0\\ &= \begin{bmatrix}p.\sigma\sum_{s=1}^{2}\xi^s(\xi^s)^\dagger & m\sum_{s=1}^{2}\xi^s(\xi^s)^\dagger\\ m\sum_{s=1}^{2}\xi^s(\xi^s)^\dagger & p.\bar{\sigma}\sum_{s=1}^{2}\xi^s(\xi^s)^\dagger\end{bmatrix}\gamma^0\\ &= \begin{bmatrix}p.\sigma\,\mathcal{I}_2 & m\,\mathcal{I}_2\\ m\,\mathcal{I}_2 & p.\bar{\sigma}\,\mathcal{I}_2\end{bmatrix}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix} = \begin{bmatrix}m\,\mathcal{I}_2 & p.\sigma\,\mathcal{I}_2\\ p.\bar{\sigma}\,\mathcal{I}_2 & m\,\mathcal{I}_2\end{bmatrix}\\ &= m\,\mathcal{I}_4 + p_0\gamma^0 + \sum_i p_i\begin{bmatrix}0 & \sigma^i\\ -\sigma^i & 0\end{bmatrix}\\ &\equiv \not{p} + m. \quad \blacksquare\end{aligned} \tag{60}$$

For Eq. (10):

$$\begin{aligned}\sum_{s=1}^{2} v^s(\vec{p})\bar{v}^s(\vec{p}) &= \sum_{s=1}^{2}\begin{bmatrix}\sqrt{p.\sigma}\xi^s\\ -\sqrt{p.\bar{\sigma}}\xi^s\end{bmatrix}\begin{bmatrix}\sqrt{p.\sigma}(\xi^s)^\dagger & -\sqrt{p.\bar{\sigma}}(\xi^s)^\dagger\end{bmatrix}\gamma^0\\ &= \begin{bmatrix}p.\sigma\sum_{s=1}^{2}\xi^s(\xi^s)^\dagger & -m\sum_{s=1}^{2}\xi^s(\xi^s)^\dagger\\ -m\sum_{s=1}^{2}\xi^s(\xi^s)^\dagger & p.\bar{\sigma}\sum_{s=1}^{2}\xi^s(\xi^s)^\dagger\end{bmatrix}\gamma^0\\ &= \begin{bmatrix}p.\sigma\,\mathcal{I}_2 & -m\,\mathcal{I}_2\\ -m\,\mathcal{I}_2 & p.\bar{\sigma}\,\mathcal{I}_2\end{bmatrix}\begin{bmatrix}0 & 1\\ 1 & 0\end{bmatrix} = \begin{bmatrix}-m\,\mathcal{I}_2 & p.\sigma\,\mathcal{I}_2\\ p.\bar{\sigma}\,\mathcal{I}_2 & -m\,\mathcal{I}_2\end{bmatrix}\\ &= -m\,\mathcal{I}_4 + p_0\gamma^0 + \sum_i p_i\begin{bmatrix}0 & \sigma^i\\ -\sigma^i & 0\end{bmatrix}\\ &\equiv \not{p} - m. \quad \blacksquare\end{aligned} \tag{61}$$

6.

$$\begin{aligned}\{\psi_\alpha(\vec{x}),\psi_\beta(\vec{y})\} = \int \frac{d^3p\,d^3q}{(2\pi)^6}\frac{1}{2\sqrt{E_{\vec{p}}E_{\vec{q}}}}\sum_{r,s=1}^{2}\{b^r_{\vec{p}}u^r_\alpha(\vec{p})e^{i\vec{p}.\vec{x}} + (c^r_{\vec{p}})^\dagger v^r_\alpha(\vec{p})e^{-i\vec{p}.\vec{x}},\\ b^s_{\vec{q}}u^s_\beta(\vec{q})e^{i\vec{q}.\vec{y}} + (c^s_{\vec{q}})^\dagger v^s_\beta(\vec{q})e^{-i\vec{q}.\vec{y}}\} = 0. \quad \blacksquare\end{aligned} \tag{62}$$

$$\begin{aligned}\{\psi^\dagger_\alpha(\vec{x}),\psi^\dagger_\beta(\vec{y})\} = \int \frac{d^3p\,d^3q}{(2\pi)^6}\frac{1}{2\sqrt{E_{\vec{p}}E_{\vec{q}}}}\sum_{r,s=1}^{2}\{(b^r_{\vec{p}})^\dagger(u^r_\alpha(\vec{p}))^\dagger e^{-i\vec{p}.\vec{x}} + c^r_{\vec{p}}(v^r_\alpha(\vec{p}))^\dagger e^{i\vec{p}.\vec{x}},\\ (b^s_{\vec{q}})^\dagger(u^s_\beta(\vec{q}))^\dagger e^{-i\vec{q}.\vec{y}} + c^s_{\vec{q}}(v^s_\beta(\vec{q}))^\dagger e^{i\vec{q}.\vec{y}}\} = 0. \quad \blacksquare\end{aligned} \tag{63}$$

$$\{\psi_\alpha(\vec{x}),\psi^\dagger_\beta(\vec{y})\} = \int \frac{d^3p\,d^3q}{(2\pi)^6}\frac{1}{2\sqrt{E_{\vec{p}}E_{\vec{q}}}}\sum_{r,s=1}^{2}\{b^r_{\vec{p}}u^r_\alpha(\vec{p})e^{i\vec{p}.\vec{x}} + (c^r_{\vec{p}})^\dagger v^r_\alpha(\vec{p})e^{-i\vec{p}.\vec{x}},$$

$$(b^s_{\vec{q}})^\dagger(u^s_\beta(\vec{q}))^\dagger e^{-i\vec{q}.\vec{y}} + c^s_{\vec{q}}(v^s_\beta(\vec{q}))^\dagger e^{i\vec{q}.\vec{y}}\} = \int \frac{d^3p\,d^3q}{(2\pi)^6}\frac{1}{2\sqrt{E_{\vec{p}}E_{\vec{q}}}}\sum_{r,s=1}^{2}$$

$$(\{b^r_{\vec{p}},(b^s_{\vec{q}})^\dagger\}u^r_\alpha(\vec{p})(u^s_\beta(\vec{q}))^\dagger e^{i(\vec{p}.\vec{x}-\vec{q}.\vec{y})} + \{(c^r_{\vec{p}})^\dagger, c^s_{\vec{q}}\}v^r_\alpha(\vec{p})(v^s_\beta(\vec{q}))^\dagger e^{-i(\vec{p}.\vec{x}-\vec{q}.\vec{y})})$$

$$= \int \frac{d^3p}{(2\pi)^3}\frac{1}{2E_{\vec{p}}}[(\sum_{r=1}^{2}u^r(\vec{p})\bar{u}^r(\vec{p})e^{i\vec{p}.(\vec{x}-\vec{y})} + \sum_{r=1}^{2}v^r(\vec{p})\bar{v}^r(\vec{p})e^{-i\vec{p}.(\vec{x}-\vec{y})})\gamma^0]_{\alpha\beta}$$

$$= \int \frac{d^3p}{(2\pi)^3}\frac{1}{2E_{\vec{p}}}[((\not{p}+m)e^{i\vec{p}.(\vec{x}-\vec{y})} + (\not{p}-m)e^{-i\vec{p}.(\vec{x}-\vec{y})})\gamma^0]_{\alpha\beta}$$

$$= [(i\partial_j\gamma^j + m)\gamma^0]_{\alpha\beta}\int \frac{d^3p}{(2\pi)^3}\frac{1}{2E_{\vec{p}}}\left(e^{i\vec{p}.(\vec{x}-\vec{y})} - e^{-i\vec{p}.(\vec{x}-\vec{y})}\right)$$

$$+ [\mathcal{I}_4\delta^{(3)(\vec{x}-\vec{y})}]_{\alpha\beta} = \delta_{\alpha\beta}\delta^{(3)(\vec{x}-\vec{y})}. \quad \blacksquare \tag{64}$$

In the last expression we used the fact that going to a coordinate system s.t. $\vec{p}\to-\vec{p}$ flips the sign of the integral, but this is a Lorentz invariant integral, hence it should be 0.
Important subtlety:

$$\partial_i e^{i\vec{p}.\vec{x}} = \frac{\partial}{\partial x^j}e^{ip^jx^j} = ip^ie^{i\vec{p}.\vec{x}} = -ip_ie^{i\vec{p}.\vec{x}} \tag{65}$$

7.

$$H = \int d^3x\,\bar{\psi}(-i\gamma^i\partial_i + m)\psi(\vec{x}) = \int d^3x\frac{d^3q}{(2\pi)^3}\frac{1}{\sqrt{2E_{\vec{q}}}}\,\bar{\psi}(-i\gamma^i\partial_i + m)$$

$$\sum_{s=1}^{2}\left(b^s_{\vec{q}}u^s(\vec{q})e^{i\vec{q}.\vec{x}} + (c^s_{\vec{q}})^\dagger v^s(\vec{q})e^{-i\vec{q}.\vec{x}}\right) = \int d^3x\frac{d^3q}{(2\pi)^3}\frac{1}{\sqrt{2E_{\vec{q}}}}\,\bar{\psi} \tag{66}$$

$$\sum_{s=1}^{2}\left(b^s_{\vec{q}}(-\gamma^iq_i + m)u^s(\vec{q})e^{i\vec{q}.\vec{x}} + (c^s_{\vec{q}})^\dagger(\gamma^iq_i + m)v^s(\vec{q})e^{-i\vec{q}.\vec{x}}\right)$$

From the Dirac equation:

$$(\not{p}-m)u(\vec{p}) = 0 \implies \gamma^0E_p\,u(\vec{p}) = (-\gamma^ip_i + m)u(\vec{p}) \tag{67}$$

and similarly:

$$(\not{p}+m)v(\vec{p}) = 0 \implies -\gamma^0E_p\,v(\vec{p}) = (\gamma^ip_i + m)v(\vec{p}) \tag{68}$$

therefore:

$$
\begin{aligned}
H &= \int d^3x \frac{d^3q}{(2\pi)^3}\sqrt{\frac{E_{\vec{q}}}{2}}\,\psi^\dagger(\gamma^0)^2\sum_s\left(b^s_{\vec{q}}u^s(\vec{q})e^{i\vec{q}.\vec{x}}-(c^s_{\vec{q}})^\dagger v^s(\vec{q})e^{-i\vec{q}.\vec{x}}\right)\\
&= \int \frac{d^3p\,d^3q}{(2\pi)^6}\sqrt{\frac{E_{\vec{q}}}{E_{\vec{p}}}}\frac{d^3x}{2}\sum_{r,s=1}^{2}\left((b^r_{\vec{p}})^\dagger(u^r(\vec{p}))^\dagger e^{-i\vec{p}.\vec{x}}+c^r_{\vec{p}}(v^r(\vec{p}))^\dagger e^{i\vec{p}.\vec{x}}\right)\\
&\left(b^s_{\vec{q}}u^s(\vec{q})e^{i\vec{q}.\vec{x}}-(c^s_{\vec{q}})^\dagger v^s(\vec{q})e^{-i\vec{q}.\vec{x}}\right) = \int \frac{d^3p\,d^3q}{(2\pi)^3}\sqrt{\frac{E_{\vec{q}}}{E_{\vec{p}}}}\frac{1}{2}\sum_{r,s=1}^{2}\\
&\left((b^r_{\vec{p}})^\dagger b^s_{\vec{q}}(u^r(\vec{p}))^\dagger u^s(\vec{q})\delta(\vec{p}-\vec{q})+c^r_{\vec{p}}b^s_{\vec{q}}(v^r(\vec{p}))^\dagger u^s(\vec{q})\delta(\vec{p}+\vec{q})\right.\\
&\left.-(b^r_{\vec{p}})^\dagger(c^s_{\vec{q}})^\dagger(u^r(\vec{p}))^\dagger v^s(\vec{q})\delta(\vec{p}+\vec{q})-c^r_{\vec{p}}(c^s_{\vec{q}})^\dagger(v^r(\vec{p}))^\dagger v^s(\vec{q})\delta(\vec{p}-\vec{q})\right)\\
&= \int \frac{d^3p}{(2\pi)^3}\frac{1}{2}\sum_{r,s=1}^{2}\left(2E_{\vec{p}}(b^r_{\vec{p}})^\dagger b^s_{\vec{p}}\delta^{rs}-2E_{\vec{p}}c^r_{\vec{p}}(c^s_{\vec{p}})^\dagger\delta^{rs}\right)\\
&= \int \frac{d^3p}{(2\pi)^3}E_{\vec{p}}\sum_{s=1}^{2}\left((b^r_{\vec{p}})^\dagger b^s_{\vec{p}}+(c^s_{\vec{p}})^\dagger c^r_{\vec{p}}\right). \quad \blacksquare
\end{aligned}
\tag{69}
$$

8.

Given Eq. (17) from the Q's file, we can prove Eq. (18) from the Q's file, by replacing {} anti-commutator with [] commutator and do nothing else. So that one is trivial. In Q.7's solution above, we used the fermionic anti-commutator only in the last step, if instead we use the bosonic commutator, we do not get the minus sign and hence:

$$
H = \int \frac{d^3p}{(2\pi)^3}E_{\vec{p}}\sum_{s=1}^{2}\left((b^r_{\vec{p}})^\dagger b^s_{\vec{p}}-(c^s_{\vec{p}})^\dagger c^r_{\vec{p}}\right). \quad \blacksquare \tag{70}
$$

Which means we can lower the energy indefinitely by having infinitely many c-particles.